



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Turing machines

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All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 165 - 175 (essential)

Turing machines

Alan Turing (1912-1954) - British Mathematician

- Formalized the abstract notion of computation that we still use today.
- Is regarded as the founding father of AI.
- Worked at Bletchly Park during the second world war and contributed to breaking german encryption.
- At Manchester University contributed to the development of the early Manchester computers in the group of Max Newman.
- Was prosecuted by the British Government for being homosexual.
- Is depicted on the 50 pound note.



Formal definition of Turing machines

Definition (Turing machine (TM))

A **Turing machine** is a 7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$, where

1. Q is the set of states,
2. Σ is the input alphabet not containing the blank symbol $\sqcup$,
3. Γ is the tape alphabet, where $\sqcup \in \Gamma$ and $\Sigma \subseteq \Gamma$,
4. $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$ is the transition function,
5. $q_0 \in Q$ is the start state,
6. $q_{\text{accept}} \in Q$ is the accept state, and
7. $q_{\text{reject}} \in Q$ is the reject state, where $q_{\text{accept}} \neq q_{\text{reject}}$.

A **configuration** of a Turing machine consists of the current state, the content of the tape, and the position of the head of the tape.

For strings $u, v \in \Gamma^*$ and $q \in Q$ we write $u q v$ for the configuration where q is the current state, $uv_{\sqcup\sqcup}\dots$ is the content of the tape and the head is positioned over the first symbol of v .

Let $a, b, c \in \Gamma$, $u, v \in \Gamma^*$ and $p, q \in Q$. We say that

- configuration $ua p bv$ **yields** configuration $u q acv$ if $\delta(p, b) = (q, c, L)$ ¹.
- configuration $ua p bv$ **yields** configuration $uac q v$ if $\delta(p, b) = (q, c, R)$ ¹.
- configuration $p bv$ **yields** configuration $q cv$ if $\delta(p, b) = (q, c, L)$.
- configuration $p bv$ **yields** configuration $c q v$ if $\delta(p, b) = (q, c, R)$.

¹Note that configuration $ua p$ is the same as configuration $ua p \sqcup$

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ be a Turing machine and let $w \in \Sigma^*$ a string. The TM M **accepts** (**rejects**) w if a sequence of configurations $C_1, \dots, C_k$ exists such that:

1. $C_1 = q_0 w$ is the starting configuration;
2. for $1 \leq i < k$ we have that C_i yields C_{i+1} ;
3. $Q_k = u q_{\text{accept}} v$ ($Q_k = u q_{\text{reject}} v$) for some $u, v \in \Gamma^*$ is an accepting (rejecting) configuration.

We let $L(M)$ be the set of all strings accepted by M .